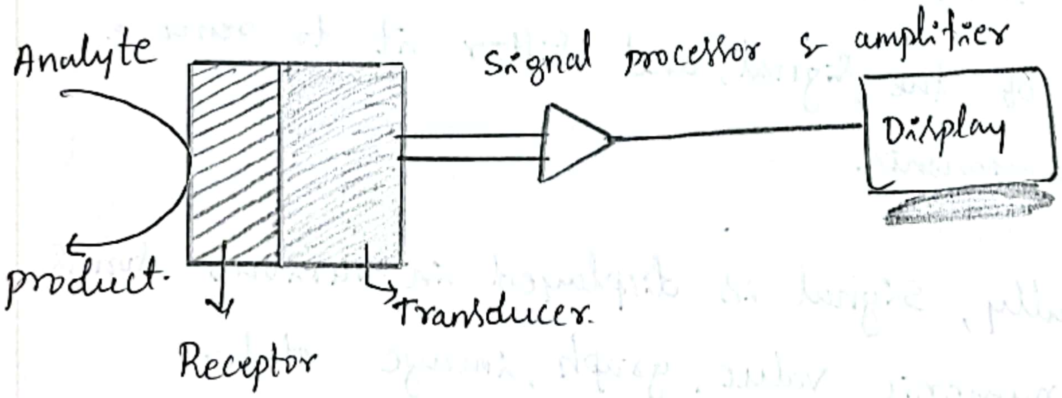
Module -1 : Sensors & Energy Systems.

Sensors

A Sensor is a device that measures the specific parameter of a physical environment for analytical information and converts it into data that can be interpreted by either a human or a machine. The physical entity is called test sample or analyte. The specific sensor interacts with the analyte in a specific manner to produce signal, which is propotional to analyte concentration. The typical components of sensor is as follows,



Receptor: Receptor is a detection component of a sensor. It interacts with the analyte in a specific and selective manner. Identifies, records, or indicates a change in one of the variables in its environment such as pressure, temperature, pH @ light intensity.

Transducer: A transducer converts the chemical @ physical quantities indicated by the receptor into electrical signals such as electrical charge, current, @ voltage. In most cases of chemical sensors, the receptor and the transducer are embedded in the same unit.

Signal processors: A signal processor is an electronic device that amplify the electrical signal from the detector. In addition, the signal processor may convert the signal from DC to AC (or vice-versa), change the phase of the signal, and filter it to remove unwanted components.

Display: Finally, signal is displayed in various forms such as numeric value, graph, image etc/-.

Sensors can be classified in various ways. on the basis of detection system (Receptor/Transducer), they are classified into * Electrochemical sensors.
* Thermometric sensors &
* optical sensors etc/-.

Electrochemical Sensors:

The electrodes are used as transducers in electrochemical sensors. Hence, transducer of an electrochemical sensor consists of sensing electrode (working), counter electrode, reference electrode, and an electrolyte. The analyte undergoes either oxidation @ reduction on the chemically modified surface of the sensing electrode. The electrolyte helps in completion of electric circuit and transportation of reactants and products. It should be stable under all conditions of sensor operation. The following steps are involved in the working of an electrochemical sensor.

1) Diffusion of the analyte to the electrode/electrolyte interface.
2). Adsorption onto the electrode surface
3) Electrochemical reaction with electron transfer
4) Desorption of the products
5) Diffusion of the products away from the reaction zone to the bulk of electrolyte @ gas phase.

Applications of electrochemical sensors

Electrochemical sensors are used in various applications in different configurations.

* The oxygen sensor is used for detection and monitoring of DO in water boiler, metallurgy, glass manufacturing H_2 fuel production etc-.

* In the detection of toxic gases, warfare agents etc- in security and defense applications.

* In water analysis and environmental monitoring, like measurement of toxic metal concentration in water, detection of oxides of NO_x, SO_x, CO, pH etc-

* In clinical diagnostic and health care applications, like in situ monitoring of glucose serum uric acid, blood Ca^{+2}, Fe^{+2}, acetylocholine etc-

* In soil analysis.

Electrochemical sensors are divided into several types,

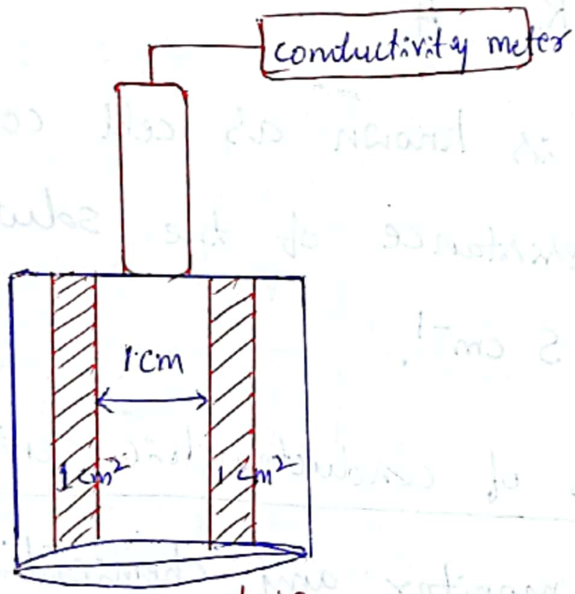
1) potentiometric (measure of voltage) sensor,

2) Ampherometric (measure of current) sensor.

3) Conductometric (measure of conductivity) sensor.

Conductometric Sensor

Electrode used in conductivity sensor is called as conductivity cell. The conductivitivity cell is dipped in the electrolytic solution taken in a beaker and it is connected to a conductance measuring device called as conductivity meter.



conductometric sensor.

conductivity sensor is used to measure the change in electrolytic conductance of the solution during replacement of ions of a particular conductivity by ions of different conductivity. The conductivity cell is made of two platinum foils with unit cross sectional area and unit distance between them. volume between two electrodes is 1 cm^3. conductance of unit volume of the solution is called as specific conduct~~ance~~ance. There will be change in specific conductance of solution when there is change in number of ions @ type of ion with different mobility. Specific conductance (K) is given by,

$$K = \frac{1}{R} \times \frac{l}{A}$$

where $\frac{l}{A}$ is known as cell constant and "R" is the resistance of the solution. It has a unit of $S\ cm^{-1}$.

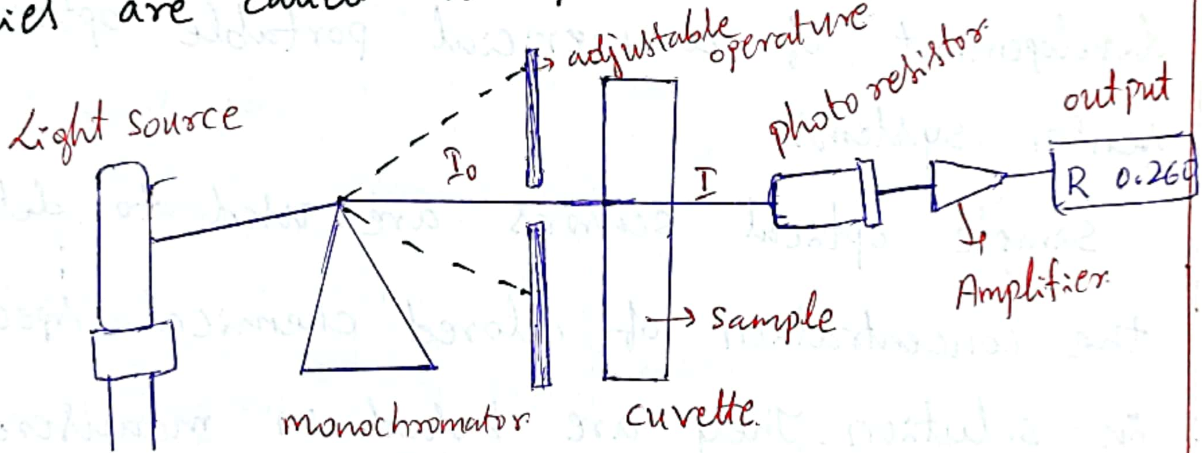
Applications of conductometric sensors:

* used to monitor any chemical which can change the electrolytic conductance of solution on chemical reaction.
* used to estimate acids, bases and their mixture in a sample
* To check the amount of ionic impurities in water samples.
* In measuring acidity @ alkalanity of sea water and fresh water.

* conductometric biosensors are used in biomedicine, environment monitoring, biotechnology and agriculture related applications.

Optical sensor

Sensors based on the transduction of interaction of electromagnetic radiation with the chemical species are called as optical sensors.



The optical signals arises from the interaction of the analyte with an incident radiation. The interaction could result in absorption, emission, scattering @ reflection of light. The type of interaction depends on the wavelength of the probing radiation and on the structure of the molecules in the analyte. The intensity of the radiation eliminating from the analyte carries information on the concentration of the analyte. It is measured by the optoelectronic instrumentation.

The main components are a light source, a wavelength selector, a photodetector, and a display of the output.

The advantage of optical ~~sensors~~ fibers and solid state optoelectronic components enabled the developement of commericial portable optical sensor systems.

Simple optical sensors are used to determine the concentration of colored chemical species in solution. They are based on meausrement of absorbance of visible light of particuliar wavelength by colored chemical species in the solution. They are governed by Beer - Lamberts law.

Applications of optical sensors:

* Determination of any chemical species which can interact with electromagnetic radiation.

  Ex: Ions in solutions (metal ions, anions), gases, (CO_2, O_2, NH_3, SO_2, NO_2 etc), vapors (moisture, volatile organic compounds etc.) and molecules (glucose, pesticides, DNA, bacteria etc.) can be determined using optical sensors.

Thermometric Sensors

Thermometric sensor is based on the measurement of thermal changes during the interaction between analyte and receptor. These are applicable to catalytic expthermic reactions.

Main component of a thermometric sensor is a small tubular catalytic reactor fitted with a temperature transducer. Analyte (reactant) is fed into the reactor. The wall of the reactor is coated with a catalyst or enzyme capable of catalysing the reaction, liberating heat energy. Heat liberated is quantified by means of a temperature transducer. The change in temperature is converted to the output voltage by tansducer which is amplified

and fed to the data storage and processing unit.
There are several types of thermal transducer which is amplified and fed to the data storage and processing unit.
There are several types of thermal transducer, two of them are discussed as follows,

1) Thermistor: It is a positive component whose resistance changes as the temperature in a system changes. The two general categories of thermistors are negative temperature co-efficient (NTC) and positive temperature co-efficient (PTC) with the NTC version being the most commonly used type. NTC thermistors are made of a ceramic semiconductor material. With NTC thermistors, resistance decreases as the temperature increases. when a component heats up, electrons are loosened from the lattice ~~Point~~. atoms. They leave and transport electricity easily. As the temperature increases, a thermistor moves electricity quickly and efficiently. The decrease in resistance is converted to output voltage using a wheatstone bridge resistor.

2) <u>Thermocouple</u>: A thermocouple a device that converts the temperature difference directly into an electrical voltage. It consists of a loop formed by two different materials (metals or semiconducoctors). The output voltage is proportional to the temperature difference between the two junctions.

<u>Applications of a thermometric sensors</u>

~~* Thermal biosensors are based on the temperature difference directly into an electrical ~~vol~~ voltage. It consists of a loop formed by two different materials ~~(metals or~~~~

* Thermal biosensors are based on the temperature change induced by a simple enzymatic reation. They are used in the determination of metabolities, bioprocess monitoring, and environmental control.

* Thermometric chemicals are used for determination of combustible gases that react with oxygen at the surface of a suitable catalyst.

Electrochemical Sensors for measurement of Dissolved oxygen (DO):

Anode : Zn/Pb , cathode (inert) : Ag , Electrolyte : NaCl.

The difference in potential between the anode and the cathode should be more than 0.5 V to reduce dissolved oxygen without an external applied potential.

When the electrode assembly is dipped in water to measure its DO, the anodic material undergoes oxidation liberating electrons.

$$2\,Zn \longrightarrow 2\,Zn^{+2} + 4e^{\ominus}$$

At the cathode, Dissolved oxygen undergoes reduction. 'Ag' cathode is inert, it only passes electrons to oxygen for reduction.

$$O_2 + 2H_2O + 4e^{\ominus} \longrightarrow 4\,OH^{\ominus}$$

overall reaction is,

$$Zn + O_2 + 2H_2O \longrightarrow 2\,Zn(OH)_2$$

The current produced by the reduction of oxygen at cathode is proportional to the partial pressure of oxygen in the water sample.
The Zinc hydroxide precipitates out and will gradually affect the sensor's performance.
This requires periodic replacement of electrolyte and anode as well as calibration.

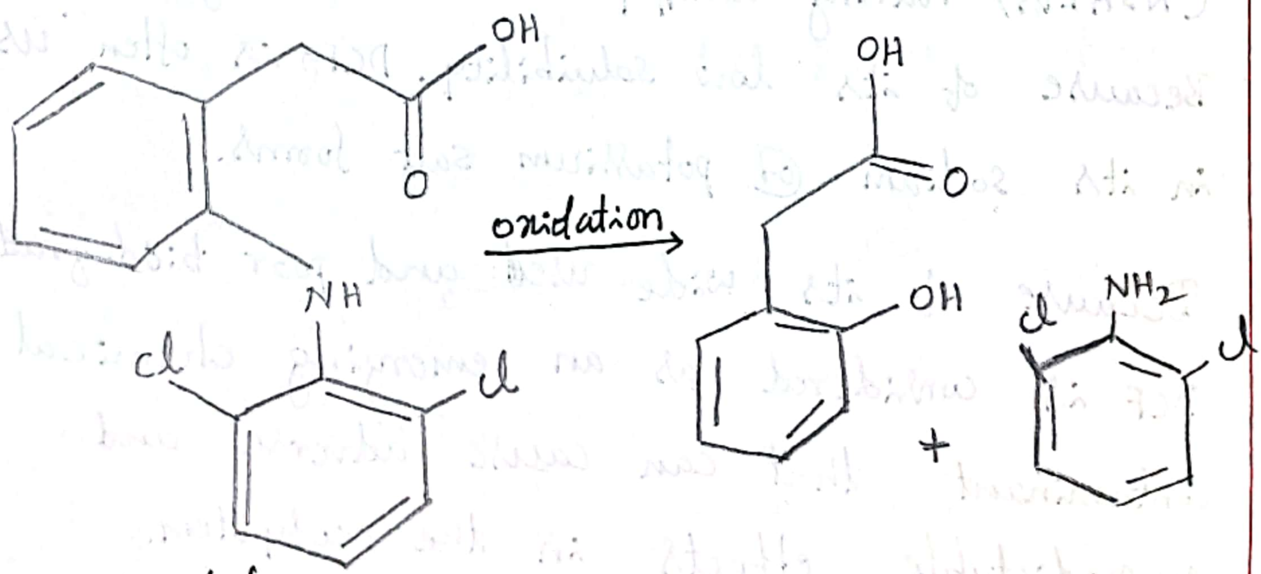
Electrochemical Sensor for Pharmaceuticals.
[Ex: Diclofenac]

Diclofenac (DCF), designated as 2-(2-((2,6 dichlorophenyl) amino)phenyl) acetic acid, is one of the widely prescribed non-steroidal anti-inflammatory drugs (NSAIDs) having antipyretic and analgesic properties. Because of its low solubility. DCF is often used in its sodium @ potassium salt forms.

Because of its wide used and poor biodegradation, DCF is considered as an emerging chemical contaminant that can cause adverse and unpredictable effects in the ecosystem.

Due to the high electroactivity of DCF, it can be detected at low concentrations using various types of potentiometric sensors.

The sensing (working) electrode is graphite carbon coated with Multi walled carbon Nanotubes (MWCNT) and gold nano particles. In the detection, along with sensing electrode, counter electrode and reference electrodes are used. When the sample containing diclofenac is put in the sensor, the following oxidation reaction of diclofenac occurs on the surface of the sensing electrode. The change in potential of the reaction gives the concentration of diclofenac.

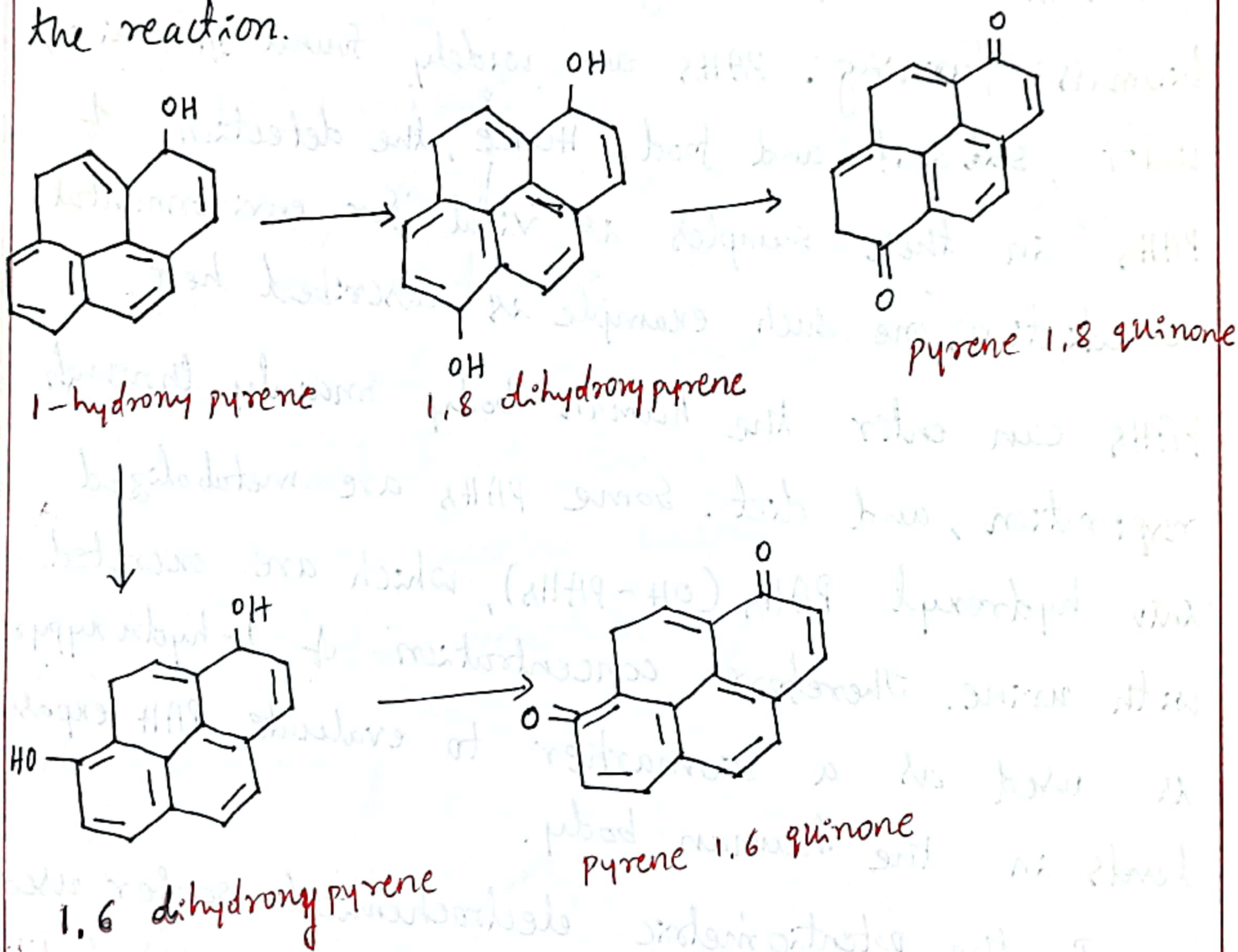


Diclofenac

Electrochemical Sensors for Hydrocarbons

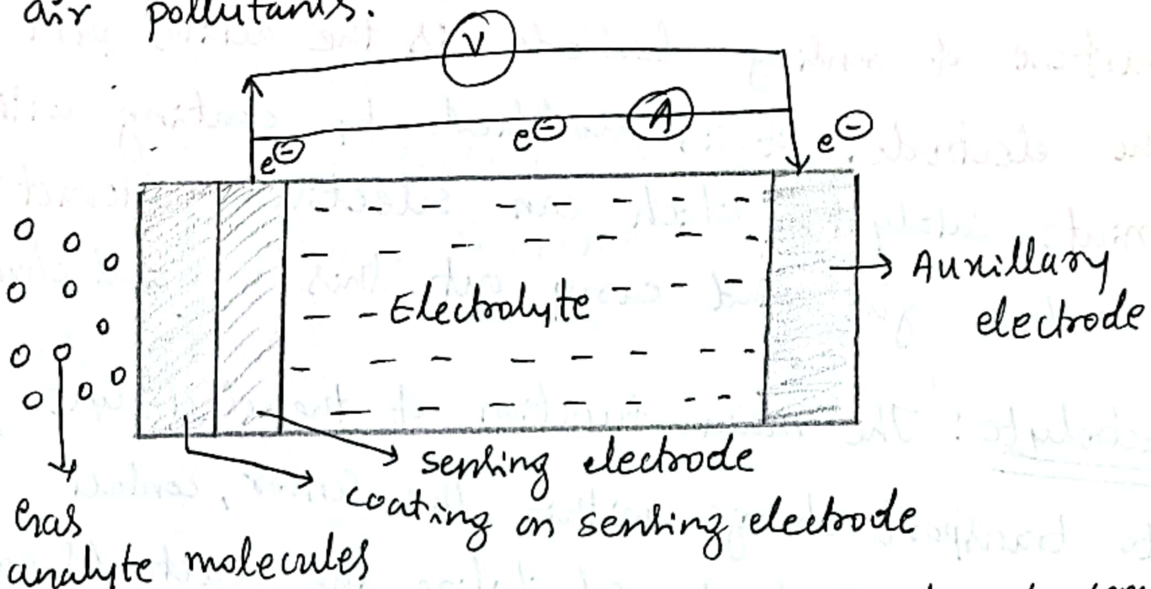
The hydrocarbon pollutants, particularly polycyclic aromatic hydrocarbons (PAHs) are known carcinogenic and mutagenic compounds. The main anthoropogenic sources are fossil fuel processing and utilization as well as biomass burning. PAHs are widely found in air, water, ~~Soil~~ Soil and food. Hence, the detection of PAHs in these samples is vital for environmental remediation. one such example is described here, PAHs can enter the human body mainly through respiration, and diet. Some PAHs are metabolized into hydroxyl PAHs (OH-PAHs), which are excreted with urine. Therefore concentration of 1-hydroxypyrene is used as a biomarker to evaluate PAH exposure levels in the human body.

In the potentiometric electrochemical sensor used to detect 1-hydroxypyrene, the sensing (working) electrode is graphite carbon coated with chromium containing metal organic framework (Cr-MOF) and graphene oxide. In addition, counter electrode and reference electrodes are used to complete the circuit and allow charge to flow. The 1-hydroxypyrene structure contains electrochemically active hydroxyl groups, which undergo oxidation at the surface of sensing electrode. concentration of 1-hydroxypyrene is determined from the change in potential of the reaction.



Electrochemical gas sensors:

Electrochemical gas sensors are mainly used to monitor the air quality by measuring the concentration of air pollutants.



electrochemical gas sensor consists of 4 components,

1) Filter: The filter is used to prevent unwanted contaminants, mainly particulate mater from entering into sensor.

2) Membrane: A gas-permeable membrane selectively allows only the gaseous analyte molecules to pass and functions to regulate the gas flow and to prevent leakage of the electrolyte from the interior of the sensor.

3) **Electrodes**: Two @ Three electrode system is used.
oxidation of analyte on the surface of sensing
electrode, liberating electrons. Transfer of liberated
electrons from anode to cathode through external circuit.
The surface of sensing electrode is the active part
of the electrode. It is modified by coating with
appropriate catalysts which can selectively interact
with analyte gas and carry out this chemical change.

4) **Electrolyte**: The main function of the electrolyte
is to transport charge within the sensor, contact
all electrodes effectively, soluibilize the reactants and
products for efficient transport. It should be a good
ionic conductor, ~~and~~ chemically & physically
stable under operation conditions of sensor.

Electrochemical gas sensors for SO_x & NO_x

Any gaseous compound which can undergo redox
reaction on the surface of electrode can be
measured with an electrochemical sensor.
Design of the sensor and composition of sensing electrode
for each gas can be unique but the components and

working principle is same as discussed above and in previous section of electrochemical sensor.

Detection of NO_2:

Detection of NO_2 in an amperometric gas sensor in aqueous electrolyte is based on the following electrochemical reduction reaction on the surface of sensing electrode. Au, pt/Nafion as sensing electrode and 10 M H_2SO_4 as electrolyte are used.

$$NO_2 + 2H^{\oplus} + 2e^{\ominus} \longrightarrow NO + H_2O$$

Detection of NO:

Detection of NO in an amperometric gas sensor in aqueous electrolyte is based on the following electrochemical oxidation reaction on the surface of sensing electrode. Au/NASICON - $NaNO_2$ is used as sensing electrode and electrolyte.

sensing electrode and electrolyte.

$$NO + 2H_2O \longrightarrow NO_3^{-2} + 4H^{\oplus} + 3e^{\ominus}$$

Detection of SO_2: In an amperometric gas sensor in aqueous electrolyte is based on the following electrochemical oxidation reaction on the surface of sensing electrode. Au/Nafion sensing electrode

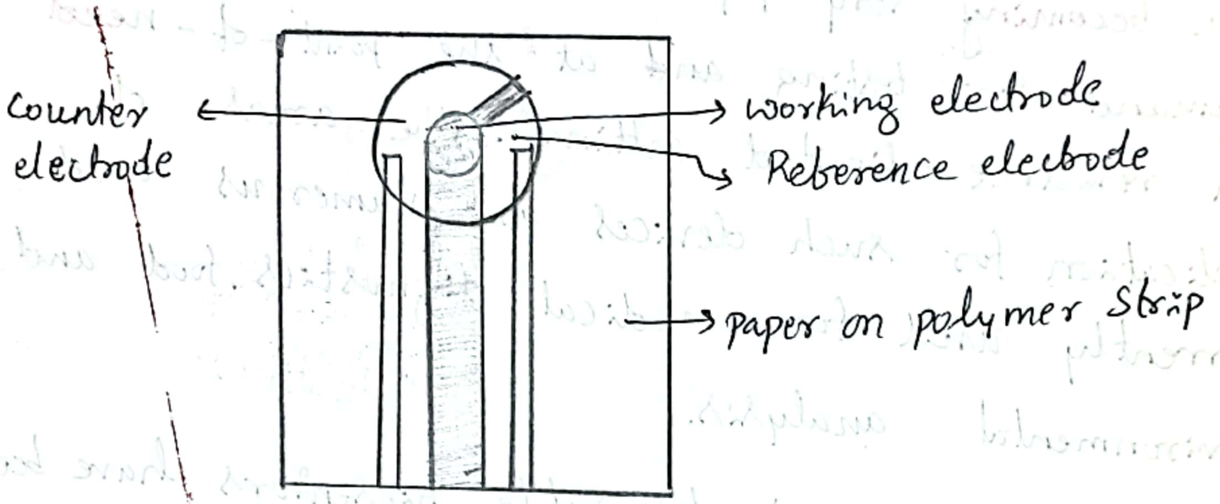
Disposable Sensors:

Disposable sensors are affordable and easy-to-use devices for short term or single-shot measurements. These sensors do not entail the risk of cross-contamination and recalibration. Disposable sensors are becoming very popular because of the increasing demand for testing ~~and~~ at the point-of-need in resource-limited settings. The areas of application for such devices are numerous and currently used for medical dignostics, food and environmental analysis.

Several types of disposable biosensors have been developed for on spot analysis of glucose, ascorbic acid, uric acid, creatinine, lactic acid and dopamine in human beings.

Disposable strip of a biosensors is a special type of paper over which all the electrodes namely reference, working and counter electrodes ~~system~~ and a bio-receptor are screen printed on a single platform as shown in Figure. Receptor contains immobilized biomaterials like enzymes, antibodies, nucleic acids, harmones, organells on its surface.

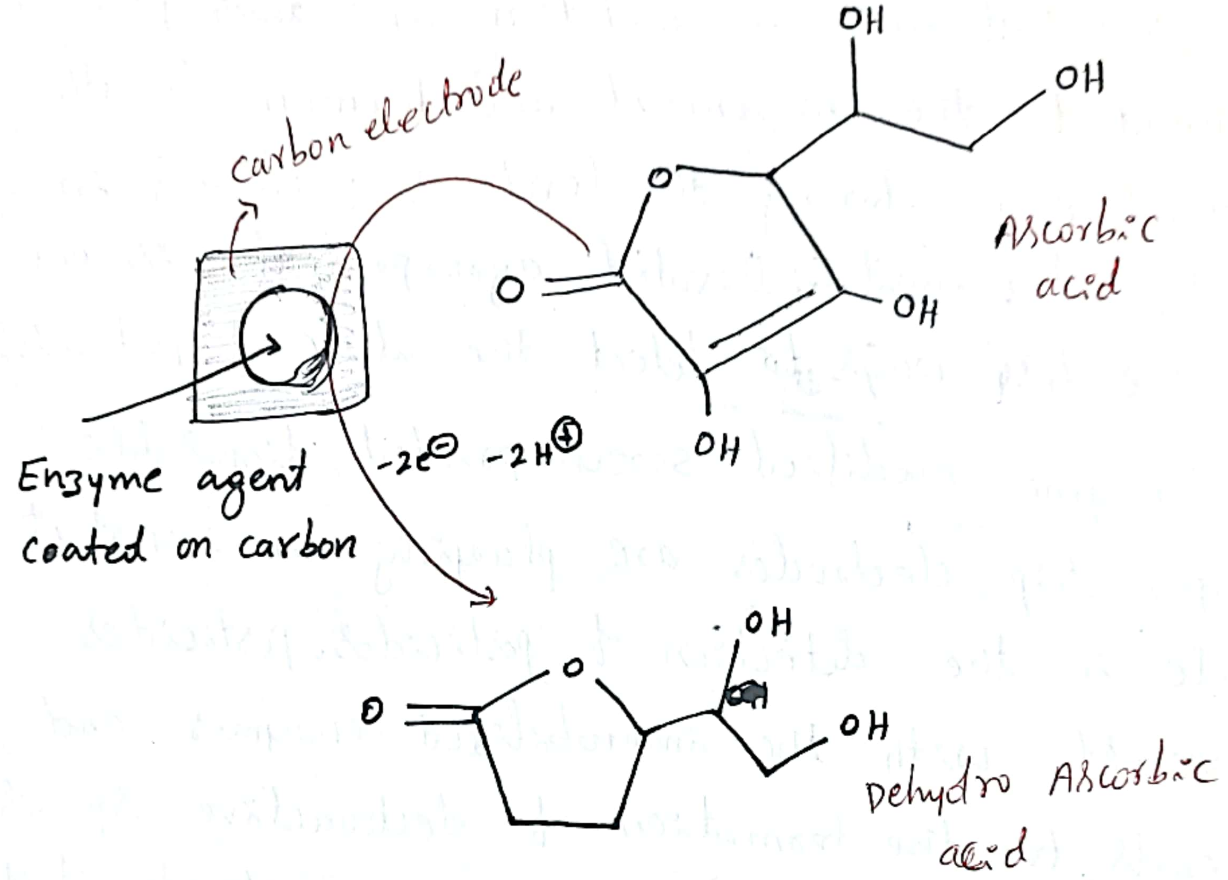
The analyte of interest will selectively bind with the biomaterial that produces the measurable electronic response. These strips can be inserted into the portable systems and used for on-site sample analysis.



Screen printed paper strip disposable Sensor.

Ascorbic acid (Vitamic-c) is a water soluable Vitamin, which helps in protecting the body against oxidative stress (antioxidant). It is abundatly available in fruits, and vegetables, in variable quantity. Hence the electrochemical biosensors which can detect ascorbic acid in various samples in lower concentration are developed. Active material coated on sensing electrode must be capable of oxidising ascorbic acid on its surface.

several active materials are being developed and one such example is given here.



The screen printed carbon electrode acts as a working electrode onto which ascorbate oxidase enzyme is immobilized with poly (ethylene glycol) and diglycidyl ether as crosslinking agents. The enzyme oxidizes the ascorbic acid in to dehydroascorbic acid. concentration of ascorbic acid is determined from the change in potential of the oxidation process.

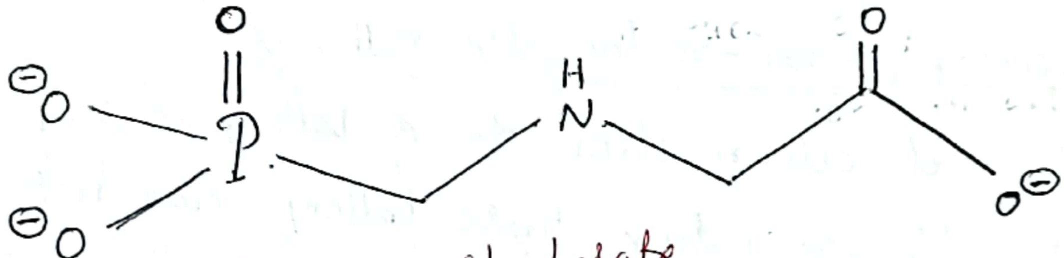
Disposable ~~electrodes~~ sensors in the detection of pesticides

The pesticides used to improve yield and protect crops result in bioaccumulation and thus pose a threat to the environment and human health. Therefore, monitoring the level of pesticides in Soil, water and cultivated agro products is one of the best ways to detect the abuse of pesticides.

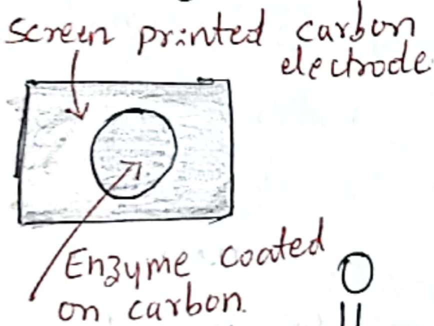
Enzyme modified screen printed disposable paper strip electrodes are playing an important role in the detection of pesticides. pesticides interact with the immobilized enzymes and leads to the formation of electroactive species. This results in decreased enzyme activity which can be measured quantitatively.

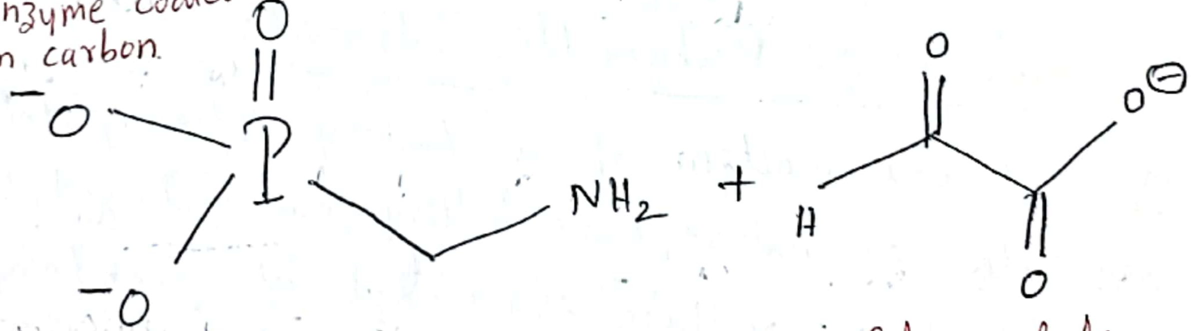
Detection of Glyphosate:

Glyphosate [N-phosphonomethyl) glycine] is an organophosphorous pesticides and its extensively used as herbicide. Glyphosate is classified as a potential carcinogenic to humans.. The glycine oxidase enzyme immobilized on a screen-printed carbon electrode can be used as sensing (working) electrode in glyphosate disposable biosensor. It oxidizes glyphosate into amidomethylphosphonic acid (CAMPA) and glyoxylate. concentration of ~~glycopho~~ glyphosate is determined from the change in potential of the oxidation process. ($\frac{1}{2}O_2$ added)



glyphosate





Aminomethylphosphonic acid (AMPA)

Glyoxylate

Energy Storage System:

Batteries

A battery is an arrangement of two @ more electrochemical cells connected either in series @ in parallel combination to achieve more voltage.

Ex: $Zn - MnO_2$ Battery, Li-Ion Battery, Sodium-ion Battery.

Batteries can be classified into three types,

1) **primary @ non-rechargable Batteries:**

The net cell reaction of a battery is completely irreversible in nature, those battery can not be rechargable are called as primary @ non-rechargable battery. Ex: $Zn - MnO_2$ Battery, $Li - MnO_2$ Battery

2) **Secondary @ Rechargable Batterries:**

The net cell reaction of a battery is completely reversible in nature, those battery can recharged to get further voltage is called as secondary @ rechargable batterries. Ex: pb storage battery, Li-ion battery.

3) Reserve Batteries: one of the component is stored seperately and is incorporated into the device at the time of application. Ex: Mg - AgCl & Mg - CuCl batteries

Lithium Ion Batterries (Construction, working and applications).

Lithium batteries have dominated the high performance non-rechargable battery market over the last 40 years.

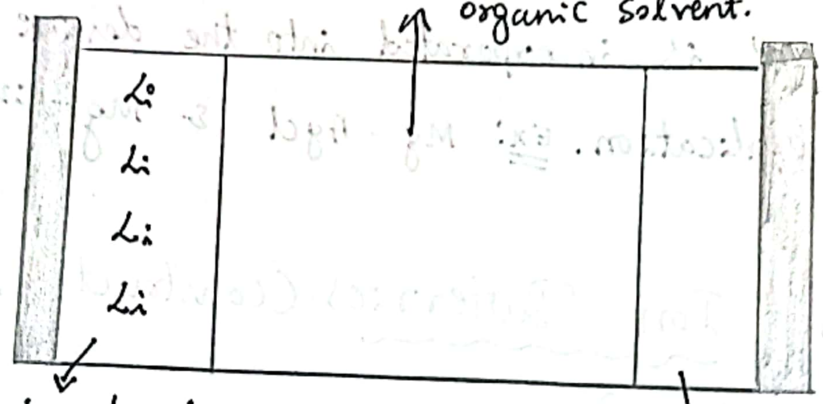
construction

The 'Li' atoms inserted into graphite carbon, acts as an anode. Lithium - metal oxide such as $LiCoO_2$ or $LiMnO_2$ acts as a cathode. A Lithium salt dissolved in binary organic solvent mixture such as $LiPF_6$ in ethylene carbonate - dimethyl carbonate behaves as an electrolyte.

The cell is represented as,

$$Li \,|\, Li^+, C \,|\, LiPF_6 \text{ in ethylene carbonate} \,|\, Li^+ - MO_2 \,|\, Li - MO_2$$

$LiPF_6$ electrolyte dissolved in organic solvent.



Anode : Li atoms intercalated in graphite layer.

cathode : Li atoms intercalated in layered structure of CoO_2 / MnO_2

Working (cell reactions)

Lithium atoms present in graphite layer (one Li atom is present for every 6 C atoms) are oxidised, liberating electrons and lithium ions. Electrons flow through external circuit ~~through~~ to cathode and lithium ions flow through the organic electrolyte towards cathode.

$$LiC_6 \xrightarrow{\text{oxidation}} Li^{+1} + 6C + e^{\ominus}$$

At cathode, Lithium ions are reduced to lithium atoms and are inserted into the layered structure of metal oxide.

$$Li + e^{\ominus} + CoO_2 \xrightarrow{\text{Reduction}} Li\text{-}CoO_2$$

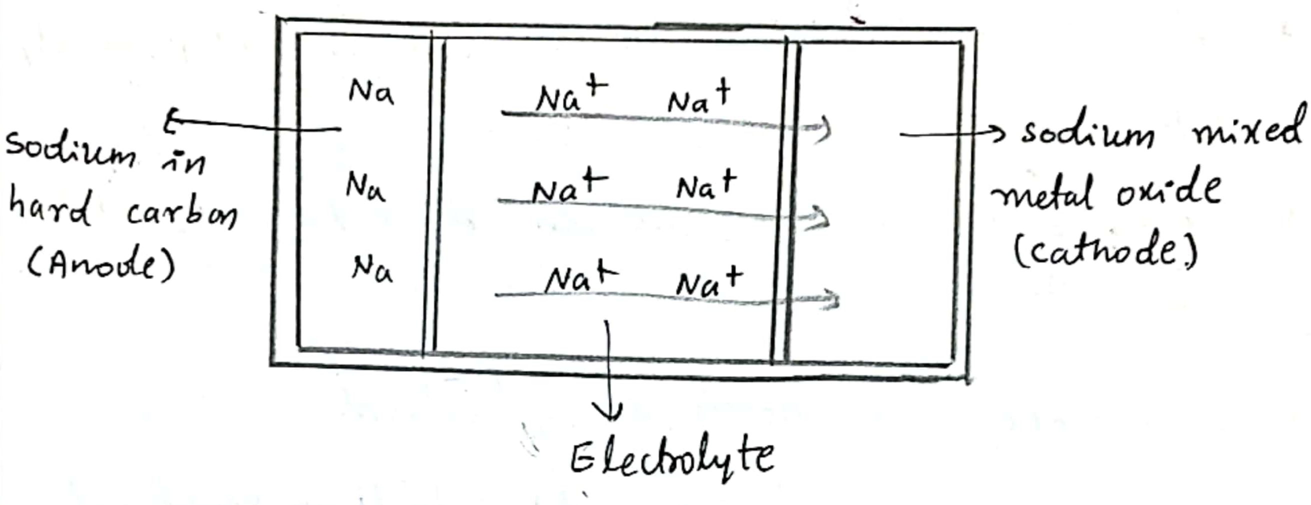
Applications:

* Li-ion batterries are used in electric vehicles.
* In portable electronic devices such as laptop, smartphone etc-.
* Li-ion batterries are used in grid-scale stationary energy storage.
* In defence and Aerospace applications
* It is used in power tools like drilling machines, cutters. etc-.

Sodium Ion Batterries:

The major challenge in Lithium ion Batterys is the limited availibility of Li source and hazardous extraction process. Hence, the supply of battery-grade Li_2CO_3 is limited. on the other hand, Na is abundant, inexpensive and ~~involves~~ involves eco-friendly extractions. Hence Sodium ion batteryes are potential future batterries.

construction



Hard carbon is used as an anode, Na - metal oxide ($Na\text{-}MO_2$), where M = Co/Mn/mixed metal oxide acts as a cathode. A sodium salt dissolved in binary organic solvent mixture such as $NaPF_6$ in ethylene carbonate - dimethyl carbonate.

working: (cell reactions)

At Anode: $C - Na \longrightarrow Na^{\oplus} + C + e^{\ominus}$

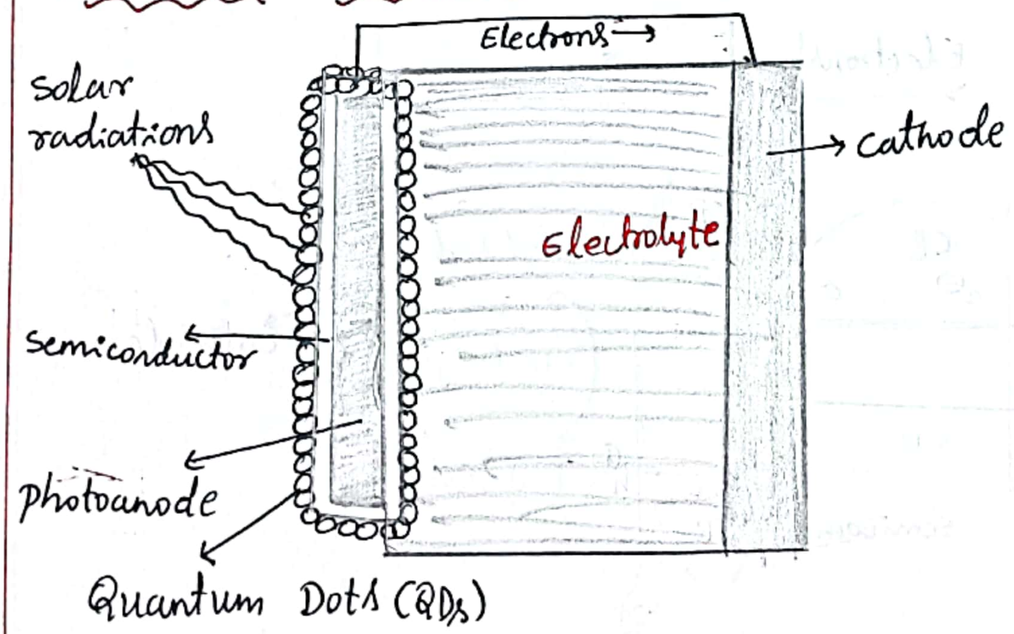
At cathode: $Na^{\oplus} + e^{\ominus} + MO_2 \longrightarrow Na - MO_2$

Quantum Dots

Quantum dots are tiny particles of a semiconducting material with a core-shell structure having diameters in the range of 2-10 nanometers (10-50 atoms). Due to their size, they exhibit unique opto-electronic properties and find application in the developement of solar photovoltaic cells to overcome the low efficiency and performance of Si based solar cells.

Ex: chalcogenides (selenides, sulfides, @ tellurides) of metals like cadmium, lead @ zinc.

Quantum Dot Sensitized Solar cells (QDSSCs)

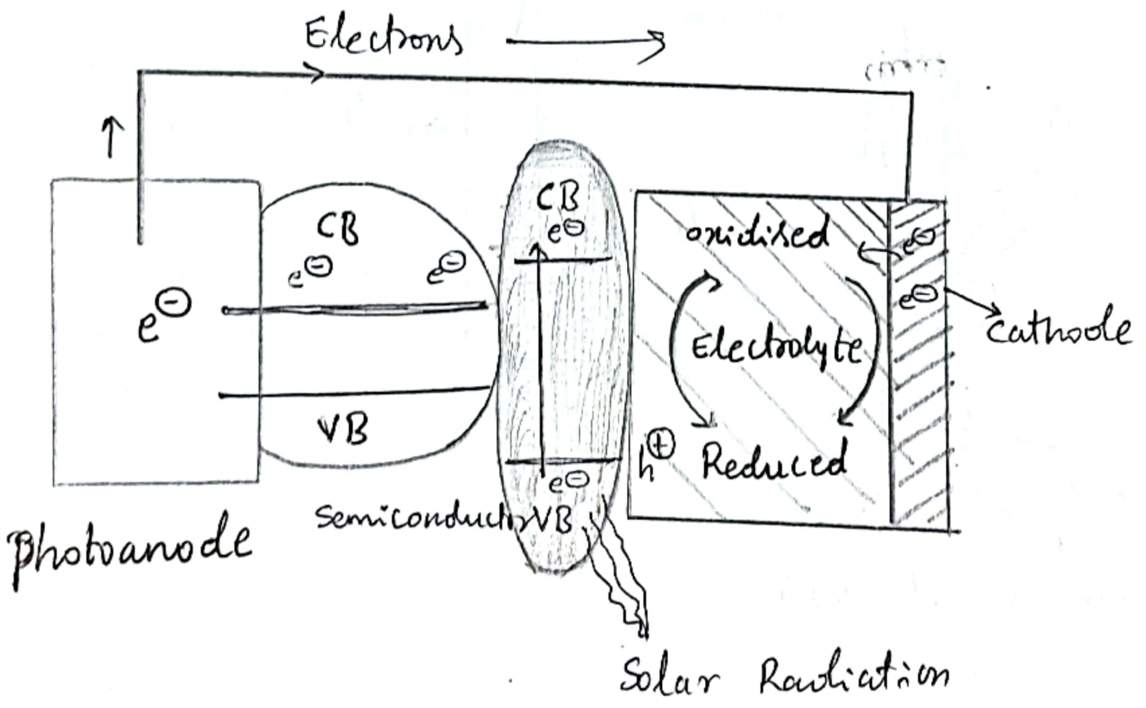


Constructions:

Photo anode: This is the working electrode. It consists of a conducting glass, over which a wide-band gap semiconductor (TiO_2, Nb_2O_5, ZnO and SnO_2) is coated. The coating is generally of with thickness of 10 mm and a porosity of 50% – 60%. Quantum dots (QDs) are deposited onto this porous layer in colloidal form.

Electrolyte: It is polysulfide material, which usually consists of a (S^{-2}/S_x^{-2}) redox couple. It is a hole conductor.

Cathode Electrode: It is generally a platinum counter electrode, which is used to regenerate electrolyte and complete the circuit.



* upon exposure to solar radiation, electron - hole pairs are formed in Quantum Dots (QDs). The electrons enter into the conduction band (CB) of the QD and holes remain in the valence band (VB).

* The injected electrons flow through a wide band gap semiconductor and glass conductor. From there it travels through the external load and completes the circuit by entering back through the cathode.

* The generated voltage is percieved as an evidence of the solar energy conversion to electric energy.

* Electrolyte takes up holes from the surface of ~~Do~~ QD's and gets reduced.

$$S^{-2} + 2h^{+} \longrightarrow S$$

$$S + S_{x-1}^{-2} \longrightarrow S_x^{-2} \quad (x = 2 \text{ to } 5)$$

* At the cathode, electrolyte is regeneragted taking up electrons from cathode.

$$S_x^{-2} \longrightarrow S + S_{x-1}^{-2}$$

$$S + 2e^{\ominus} \longrightarrow S^{-2}$$

End